

WEEK-7 SKILL

1 . Process Synchronization and Child Process Monitoring Using waitpid()

Aim : To create multiple child processes and use the `waitpid()` system call to synchronize and monitor child processes, obtain their termination status, and ensure that the parent process correctly waits for its children.

Program :

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t child1, child2;
    int status;

    printf("Parent process started.\n");
    printf("Parent PID: %d\n", getpid());

    // Create first child
    child1 = fork();
    if (child1 < 0)
    {
        perror("fork failed");
        return 1;
    }
    if (child1 == 0)
    {
        printf("\nChild 1 started.\n");
        printf("Child 1 PID: %d\n", getpid());
        sleep(3);
        printf("Child 1 completed.\n");
        exit(10);
    }

    // Create second child
    child2 = fork();
    if (child2 < 0)
    {
```

```
    if (child2 == 0)
    {
        printf("\nChild 2 started.\n");
        printf("Child 2 PID: %d\n", getpid());

        sleep(5);

        printf("Child 2 completed.\n");
        exit(20);
    }

    // Parent waits for Child 1
    printf("\nParent is waiting for Child 1...\n");

    if (waitpid(child1, &status, 0) == -1)
    {
        perror("waitpid failed");
        return 1;
    }

    if (WIFEXITED(status))
    {
        printf("Child 1 terminated normally.\n");
        printf("Child 1 exit status: %d\n", WEXITSTATUS(status));
    }

    // Parent waits for Child 2
    printf("\nParent is waiting for Child 2...\n");

    if (waitpid(child2, &status, 0) == -1)
    {
        perror("waitpid failed");
        return 1;
    }

    if (WIFEXITED(status))
    {
        printf("Child 2 terminated normally.\n");
        printf("Child 2 exit status: %d\n", WEXITSTATUS(status));
    }

    printf("\nAll child processes have completed.\n");
    printf("Parent process completed.\n");

    return 0;
}
```

Output :

```
root@keerthana: ~/waitpid_p × + ∨
root@keerthana:~/waitpid_practical# pwd
/root/waitpid_practical
root@keerthana:~/waitpid_practical# nano waitpid_demo.c
root@keerthana:~/waitpid_practical# ls
waitpid_demo.c
root@keerthana:~/waitpid_practical# gcc waitpid_demo.c -o waitpid_demo
root@keerthana:~/waitpid_practical# ls
waitpid_demo  waitpid_demo.c
root@keerthana:~/waitpid_practical# ./waitpid_demo
Parent process started.
Parent PID: 431

Child 1 started.
Child 1 PID: 432

Parent is waiting for Child 1...

Child 2 started.
Child 2 PID: 433
Child 1 completed.
Child 1 terminated normally.
Child 1 exit status: 10

Parent is waiting for Child 2...
Child 2 completed.
Child 2 terminated normally.
Child 2 exit status: 20

All child processes have completed.
Parent process completed.
root@keerthana:~/waitpid_practical# |
```

2 . To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

Aim : To understand how Linux uses the `PATH` environment variable to search for executable commands, verify executable permissions, handle missing commands, and test command resolution.

Program :

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>

int main()
{
    char *path;
    char *path_copy;
    char *dir;
    char full_path[1024];
    struct stat file_info;

    printf("==== PATH VARIABLE AND COMMAND RESOLUTION ==== \n\n");

    /* Retrieve PATH variable */
    path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable is not set.\n");
        return 1;
    }

    printf("PATH variable: \n%s\n\n", path);

    /* Make a copy because strtok modifies the string */
    path_copy = strdup(path);

    if (path_copy == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    printf("Search directories: \n");

    dir = strtok(path_copy, ":");
```

```
while (dir != NULL)
{
    printf("%s\n", dir);
    dir = strtok(NULL, ":");
}

free(path_copy);

/* Test command resolution */
printf("\n===== COMMAND RESOLUTION =====\n");

const char *command = "ls";

path_copy = strdup(path);
dir = strtok(path_copy, ":");

while (dir != NULL)
{
    snprintf(full_path, sizeof(full_path), "%s/%s", dir, command);
    if (stat(full_path, &file_info) == 0)
    {
        printf("Command '%s' found at: %s\n", command, full_path);

        if (file_info.st_mode & S_IXUSR)
        {
            printf("Executable permission: YES\n");
        }
        else
        {
            printf("Executable permission: NO\n");
        }

        break;
    }

    dir = strtok(NULL, ":");
}

if (dir == NULL)
{
    printf("Command '%s' not found in PATH.\n", command);
}

free(path_copy);

/* Test missing command */
command = "abcxyz";

printf("\nTesting missing command: %s\n", command);

path_copy = strdup(path);
```

GNU nano 7.2

```
    dir = strtok(NULL, ":");
}

if (dir == NULL)
{
    printf("Command '%s' not found in PATH.\n", command);
}

free(path_copy);

/* Test missing command */
command = "abcxyz";

printf("\nTesting missing command: %s\n", command);

path_copy = strdup(path);
dir = strtok(path_copy, ":");

int found = 0;

while (dir != NULL)
{
    snprintf(full_path, sizeof(full_path), "%s/%s", dir, command);

    if (access(full_path, X_OK) == 0)
    {
        printf("Command found at: %s\n", full_path);
        found = 1;
        break;
    }

    dir = strtok(NULL, ":");
}

if (!found)
{
    printf("Command '%s' not found in PATH.\n", command);
}

free(path_copy);

printf("\n===== TEST COMPLETED =====\n");

return 0;
}
```

Output :

```

root@keerthana: ~/path_prac X + v
root@keerthana:~/waitpid_practical# cd ~
root@keerthana:~# mkdir path_practical
root@keerthana:~# cd path_practical
root@keerthana:~/path_practical# pwd
/root/path_practical
root@keerthana:~/path_practical# nano path_resolution.c
root@keerthana:~/path_practical# ls
path_resolution.c
root@keerthana:~/path_practical# gcc path_resolution.c -o path_resolution
root@keerthana:~/path_practical# ls
path_resolution path_resolution.c
root@keerthana:~/path_practical# ./path_resolution
===== PATH VARIABLE AND COMMAND RESOLUTION =====

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common
Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Ora
cle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0/
:/mnt/c/windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Users
/sai keerthana/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/sai keerthana/AppData/Local/Programs/Python/Python314:/
mnt/c/Users/sai keerthana/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/sai keerthana/AppData/Local/Microsoft/WindowsApps:/mnt
/c/Users/sai keerthana/Downloads/38833FF26BA1D.UnigramPreview_g9c9v27vpyspw!App/gcc_15_1_0_gdb_16_3_90_20250511_binutils_2_44_mingw_w
64_v12_0_0/bin:/mnt/c/Program Files/Java/jdk-25:/mnt/c/Users/sai keerthana/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/
sai keerthana/AppData/Roaming/npm:/snap/bin

Search directories:
/usr/local/sbin
/usr/local/bin
/usr/sbin
/usr/bin
/sbin
/bin
/usr/games

```

```

root@keerthana: ~/path_prac X + v
Search directories:
/usr/local/sbin
/usr/local/bin
/usr/sbin
/usr/bin
/sbin
/bin
/usr/games
/usr/local/games
/usr/lib/wsl/lib
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/windows/system32
/mnt/c/windows
/mnt/c/windows/System32/Wbem
/mnt/c/windows/System32/WindowsPowerShell/v1.0/
/mnt/c/windows/System32/OpenSSH/
/mnt/c/Program Files/dotnet/
/mnt/c/Program Files/nodejs/
/mnt/c/Program Files/Git/cmd
/mnt/c/Users/sai keerthana/AppData/Local/Programs/Python/Python314/Scripts/
/mnt/c/Users/sai keerthana/AppData/Local/Programs/Python/Python314/
/mnt/c/Users/sai keerthana/AppData/Local/Programs/Python/Launcher/
/mnt/c/Users/sai keerthana/AppData/Local/Microsoft/WindowsApps
/mnt/c/Users/sai keerthana/Downloads/38833FF26BA1D.UnigramPreview_g9c9v27vpyspw!App/gcc_15_1_0_gdb_16_3_90_20250511_binutils_2_44_min
gw_w64_v12_0_0/bin
/mnt/c/Program Files/Java/jdk-25
/mnt/c/Users/sai keerthana/AppData/Local/Programs/Microsoft VS Code/bin
/mnt/c/Users/sai keerthana/AppData/Roaming/npm
/snap/bin

===== COMMAND RESOLUTION =====

```

```

===== COMMAND RESOLUTION =====
Command 'ls' found at: /usr/bin/ls
Executable permission: YES

Testing missing command: abcdxyz
Command 'abcdxyz' not found in PATH.

===== TEST COMPLETED =====
root@keerthana:~/path_practical# which ls
/usr/bin/ls
root@keerthana:~/path_practical# ls -l "$ (which ls)"
-rwxr-xr-x 1 root root 142312 Apr 5 2024 /usr/bin/ls
root@keerthana:~/path_practical# command -v ls
alias ls='ls --color=auto'
root@keerthana:~/path_practical# command -v pwd
pwd
root@keerthana:~/path_practical# command -v gcc
/usr/bin/gcc
root@keerthana:~/path_practical# command -v abcdxyz
root@keerthana:~/path_practical# echo $?
1
root@keerthana:~/path_practical# |

```